Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

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***Abstract*—Location-based social networks (LBSNs) record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can cache content or reserve capacity ahead of demand. Two coarse questions about the next visit are usually enough: its category (the type of place) and its region (which part of the city). These are normally handled by separate models. We therefore test whether one model can learn both tasks, and what sharing a single model costs in per-task accuracy. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across six datasets, five U.S. states and one non-U.S. city (Istanbul), this lifts next-category prediction over a standard place embedding (about $+28$ to $+40$ macro-averaged F1), and most of the gain comes from the per-visit context. We then train one multi-task model for both tasks. At every dataset, it outperforms a dedicated category model (about $+5$ to $+9$ macro-F1), and on the next-region task it outperforms the dedicated region model at four of the six datasets, while matching it (statistically, within two points) at the other two. Across the five U.S. states, the gain on region grows with the number of regions. On the non-U.S. city, the result holds: the joint model outperforms on category and is slightly ahead on region.**

***Index Terms*—mobility data, next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks**

I. Introduction

Location-based social networks, such as Gowalla [1], let people share the places that they visit as check-ins, which together record how people move through a city. Individual mobility is highly predictable in principle [2], so a service that anticipates the next move can prepare ahead of time, caching content where a user is heading [3] or provisioning capacity at the destination before demand arrives. Handover predictions already let cellular services adapt in advance at the network level [4]; at the city scale, check-in mobility analysis is likewise a design input for mobility-aware services [5].

Predicting the next place exactly, a single point of interest (POI), is hard and often more than a service needs. Two coarser questions are usually enough: the next category, the type of place such as food or shopping, and the next region, the part of the city that a user is heading to (a neighborhood-scale unit such as the U.S. census tract). The first captures intent, the second captures geography; we study whether one model should learn both at once.

Sharing one representation across tasks has a cost: in multi-task learning (MTL), one model does several jobs at once by sharing most of its parts, so the shared parameters can converge to a compromise optimal for neither task, helping one while hurting the other [6]. Prior work observed exactly this for next-category and next-region [7]; the useful question is where sharing helps, what it costs, and how to share so the gains hold and the cost stays small.

We propose two enhancements. First, we build a check-in-level representation: instead of one fixed vector per place, each check-in gets its own vector that holds its context (the time, nearby places, and recent visits). This adds a fourth, check-in level beneath the place, region, and city levels of hierarchical graph infomax [8], [9] (Fig. 1). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared trunk (a cross-attention stack where the two tasks exchange semantic context) and a private spatial path for the region task (Fig. 2). We evaluate on two settings chosen to differ: five U.S. states of different sizes (Gowalla) and one international city (Istanbul).[1]

Our contributions are as follows.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit, improving next-category prediction over a standard place embedding by about $+28$ to $+40$ points of macro-F1 (a balanced score that counts every category equally) across all six datasets (Table II). A controlled test attributes most of the gain to the per-visit context, not to extra training signal. The gap is large because a single fixed vector per place cannot separate places that serve several purposes, which per-visit context recovers.
- A single model for joint next-category and next-region prediction, sharing semantic context across tasks while keeping a private spatial path for the region task; to our knowledge, the first to treat fine-grained region as an end target of equal standing (Section II-B). In one forward pass, it outperforms a dedicated single-task category model at every dataset (by about $+5$ to $+9$ points of

[1] Code (model, representation, baselines, statistical tests): https://github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac. Both data sources are public: the category-annotated Gowalla dump (https://figshare.com/articles/dataset/gowalla_data/22126586) and Massive-STEPS [10].

macro-F1), even though each dedicated model is tuned per dataset. On region, it outperforms at four of the six datasets and stays statistically non-inferior within a two-point margin (TOST, a test of equivalence) at the remaining two (Table III).
- An empirical account, across five U.S. states and one international city, of how the joint gain scales with the number of regions. The category gain holds at all six datasets. Across the U.S. states, the region result moves monotonically with the region count: the joint model matches the dedicated model (TOST, ± 2 percentage points, or pp) at the small region counts and outperforms it at the large region counts, with the state with the most regions (California) showing the largest region gain (Fig. 3). Istanbul, the dataset with the fewest regions, is also ahead on region. California and Texas are measured at a single random initialization, the other four datasets at four (Section VI-B).

The rest of the paper presents related work (Section II), the problem (Section III), the method and setup (Sections IV and V), results (Section VI), and discussion and conclusions (Sections VII and VIII).

II. Background and Related Work

A. From place embeddings to per-visit embeddings

A place embedding transforms a POI into a vector so that similar or nearby places sit close together in the vector space. Deep Graph Infomax (DGI) [9], a general self-supervised method for graphs, learns such vectors on a place network linked by similarity, time, and distance, training them to tell the real network from a copy with shuffled node features. Hierarchical Graph Infomax (HGI) [8] adds geographic structure, organizing the network as place, then region, then city. Both produce one vector per place, so two visits to the same coffee shop look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [11], the closest example, learns a vector per visit by masking and reconstructing parts of a user's check-in sequence. Our representation, Check2HGI, also works at the check-in level but through a different construction: it keeps the hierarchical place-to-region-to-city network and the same infomax objective, now extended one level deeper, from individual places down to individual check-ins (Fig. 1), rather than switching to a sequence model as CTLE does.

The novelty is this specific combination: per-visit context inside a hierarchical graph-infomax representation. We compare against CTLE and a feature-concatenation control to separate the combination from contextualization alone and from feature injection.

B. Predicting the next category and the next region

A large body of work predicts the next place that a user will visit, the exact POI, with recurrent and attention models such as DeepMove [12], STAN [13], and GETNext [14]. In contrast, we target two properties of the next visit, its category and its region, rather than the exact place. We choose them because they are what a mobility-aware service can act on, not to make the task easier; the region alone spans hundreds to several thousand classes. Predicting over a partition of the map is also the standard formulation in the human-mobility literature, with a grid cell as the target [15]; our next-region task substitutes official neighborhood-scale units for grid cells. Our earlier work [7] established this two-task setup and observed negative transfer (sharing hurt one task); this paper adds the check-in-level representation, on which the transfer reverses (Section VI-B). The field increasingly models several granularities at once; in those systems, category and region are auxiliary signals that help a primary next-place task (MCMG [16], HMT-GRN [17]). We study the pair as the object itself. To our knowledge, fine-grained region as an end target of equal standing, rather than an auxiliary coarse grid cell, is underexplored. The nearest exceptions do not study our exact pairing: DRRGNN [18] forecasts a person's next activity region jointly with a mobility-intention label, but over regions discovered per person rather than a fixed citywide partition; a generative recommender [19] predicts category and region together, but only as auxiliary steps toward its next-place ranking.

C. Multi-task learning for mobility

MCARNN [20] jointly predicts activity and location, and a recent hierarchy-aware model continues the pattern [21]; in both, the location target is the exact place. The closest alternative to our setting is the cascade: predict the category first, then the place given the category [22]. CSLSL [23] is its current form, predicting in a chain (when, then what, then where), with location as the primary output and category as an intermediate step; CatDM [24] likewise uses the predicted category only to filter candidate places. We instead predict category and region in parallel, as end targets of equal standing in one model with a single forward pass, and we drop the next-place target entirely, so neither task is an intermediate step toward a third. A chain would have to promote one of the two targets to an upstream stage whose errors the other inherits, an order the tasks do not define. Since the cascade is the pattern that the field uses, we test the choice directly (Section VI-B). On optimization, we are conservative by design: joint training with a fixed loss weighting is standard practice [6], not itself our contribution. Reports find that gradient-balancing methods, which re-weight the tasks' updates during training (PCGrad [25], Nash-MTL [26]), rarely improve on a well-tuned fixed weighting at two tasks [27]. We confirm this: none of the balancers that we tried, including the two named above, improved on a tuned fixed task weighting in our model. The reason is visible in the gradients: the next-category and next-region updates on the shared trunk are near zero-correlated throughout training, so a balancer has no conflict to resolve, consistent with the mechanism test in Section VI-B; this is a finding for this pair of tasks, not a general rule.

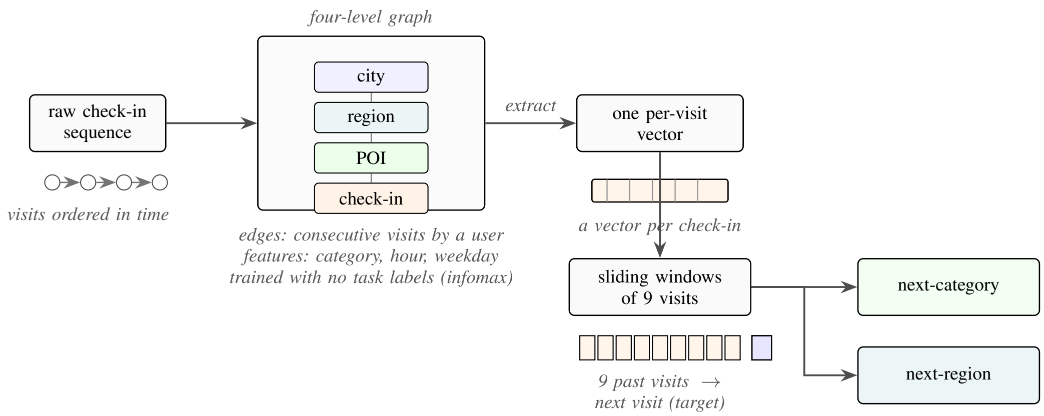

Fig. 1. From a check-in sequence to two predictions: each visit is embedded in a four-level graph (check-in, place, region, city) and described in its own context; sliding nine-visit windows feed a single model that outputs the next category and the next region.

III. Problem and Tasks

Given a user's time-ordered check-in history, we predict two properties of the next visit. The first is its *category*, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. Istanbul's source collection maps its places onto the same seven labels (Section V-A). The second is its *region*, the place's census tract (the *mahalle* for Istanbul). We do not predict the exact next place; both properties are easier to learn and, for most uses, enough. Region, unlike category, is a large label set: we frame next-region as classification over candidate regions, from 520 classes (Istanbul) to 8,501 (California; Table I).

The motivation is practical: the category says what type of place comes next, hence what a user will want; the region says where, hence where to prepare. The mobility literature supports such preparation, having long built city services on location-based social network traces [28]. A recent analysis of tourist check-in mobility finds that visits concentrate on a few hub places, argues that crowds and demand therefore concentrate, and ties this structure to infrastructure planning and adaptive strategies [5]. That analysis reads past visits; we predict two properties of the next one, and aggregated over users, those predictions point to where demand is heading before it arrives. A census tract is a neighborhood, not a radio cell, so we scope our claims to neighborhood-level preparation: demand and load anticipation, caching content ahead of time, and capacity planning; not cell association or handover. We build and evaluate no such service here; Section VII quantifies what these predictions would give such a service.

IV. Method

A. The check-in-level representation

We want each visit, not each place, to have its own vector. Figure 1 shows the full data flow, from check-ins to the two outputs.

We build a graph with four levels: the check-in, its place, the place's region (a census tract), and the city. Edges connect each level to the one above it and link a user's consecutive check-ins with a weight that decays as the time gap between the visits grows; same-place visits connect through their shared place node one level up, and nearby places are linked at the place level. Each visit's category, time of day, and day of week enter as input features of its node, not as edges. This keeps a hierarchical graph's place-to-region-to-city geography and adds the check-in as a bottom level. We train the graph mainly with an infomax objective, under which each vector learns to match its real neighborhood and reject a shuffled one [8], [9], plus two small label-free auxiliary terms: a masked reconstruction of each place's aggregated category features and an anchor to a place embedding pre-trained, label-free, on the same data (weights 0.3 and 0.1). The training uses no task label: it never sees the next category or the next region. From the trained graph, we extract one 64-dimensional vector per check-in (its region nodes have trained vectors as well), so a model sees a sequence of per-visit vectors rather than repeated per-place ones.

B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts the next visit's category and region at once. Figure 2 shows the design. Two input streams, a semantic one with the window of per-visit vectors and a spatial one in which each visit is represented by the trained vector of its region node from the same graph, pass through private per-task encoders (a small input network dedicated to each task, with no weights shared between them) into the shared trunk, a cross-attention stack of two blocks, where the two streams exchange information: in each block, attention lets each one read the other's features, while each keeps its own feed-forward weights. The tasks therefore share by exchanging information between per-task streams, not by owning hidden layers in common. The trunk then feeds a category output and a region output; the region output keeps a private spatial path, a small branch inside the one model (not a second model) that the category task does not touch.

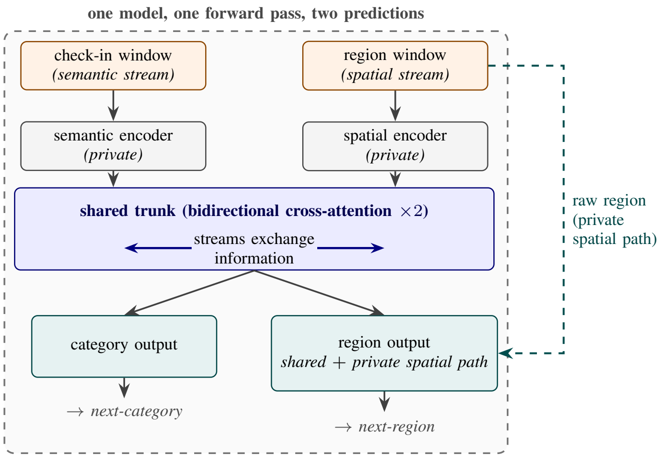

Fig. 2. The single multi-task model. A check-in (semantic) window and a region (spatial) window pass through private encoders into the shared trunk, the only place where the two tasks interact. The category output uses the shared output alone; the region output combines it with a private spatial path. One model, one forward pass, two predictions.

The training loss is a fixed-weight sum of the two outputs,

$$L = 0.75\,L_{\text{cat}} + 0.25\,L_{\text{reg}}, \tag{1}$$

where L_{cat} and L_{reg} are plain unweighted cross-entropy losses. The weights were tuned once on validation during development and held fixed across all six datasets. Class-weighting, tested on both outputs, lowered both region accuracy and category macro-F1. This is standard fixed-weight joint training [6], kept simple by design so that any improvement over the dedicated single-task models comes from the shared representation, not from an adaptive weighting scheme.

Serving both tasks this way has a cost. The joint model includes the shared trunk and both task outputs, so it is larger than either dedicated single-task model, and a forward pass costs more compute than running the two small dedicated models: the joint model has about 4.2 million parameters at Alabama against 1.1 million for the two dedicated models combined (5.2 against 2.0 at California). What the single model provides is operational rather than arithmetic: one artifact to train, version, and deploy, and one forward pass whose one set of inputs produces both answers at once.

V. Experimental Setup

A. Data

Table I summarizes our six datasets. Five come from Gowalla, a location-based social network (collected 2009 to 2011), covering the U.S. states of Alabama, Arizona, Florida, Texas, and California [1]; the sixth is Istanbul, from the Massive-STEPS collection [10], a non-U.S. check on the findings. The states range from about 114,000 check-ins and 1,109 regions in Alabama to about 3.2 million check-ins and 8,501 regions in California. Every dataset uses the same seven place categories: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. The region unit is a census tract for the Gowalla states and a mahalle (a municipal neighborhood) for Istanbul.

B. Windows, splitting, and the integrity of the representation

Windows. For each user with at least ten visits, we build time-ordered overlapping sliding windows of nine visits, one starting at each visit, with the next visit as the target, giving both tasks more examples. Because many of a user's windows would end on the same final visit, we keep one full-context window ending on that visit and drop the padded duplicates.

Splitting. We split by user with stratified five-fold cross-validation, so all of a user's windows fall in the same fold and overlap cannot leak: a test user's visits never appear in training. The held-out fold is the validation data; we reserve no third split. The split is stratified on the next-category label because the seven-class task is imbalanced: Food is the majority class in every dataset, from about 25 percent of visits in Florida to 34 percent in Alabama (Table I).

Integrity of the representation. We train the representation once on the whole dataset and feed it to every model, dedicated and joint; we verify that it holds no usable information about the test visits, for three reasons. First, its objective is label-free: it contrasts real graph neighborhoods against shuffled ones, its auxiliary terms reconstruct only input-derived quantities, and it never sees a next-category or next-region target. Second, because the graph is built over all places, the only information that can cross from the test side is the structure of the graph itself. We measured this directly: for each fold, we rebuilt the representation from that fold's training users only and re-ran both tasks against the full-corpus version. The differences are within fold noise, with no systematic gain: on region, −0.33 points at Alabama, +0.01 at Arizona, and −0.12 at Florida; on category, +0.29 at Alabama, +0.27 at Arizona, and no change (0.00) at Florida, measured on the visits whose places appear in training (about 67 percent at Alabama to 87 percent at Florida). The remaining visits are the one residual this measurement cannot reach, because the representation cannot score a place never seen in training. Third, the one component that could pass visit-to-visit information, a region-transition prior (a table of how often one region follows another), inflated region accuracy by 13 to 27 points when an earlier version built it on the whole dataset; any such table is therefore built per fold, from training data only, in every repetition of the experiment. The joint and dedicated models do not use this prior; it survives only in the HMT-GRN baseline (Section V-D). The baseline representations meet the same standard: HGI is pre-trained once on the whole dataset, like ours, and CTLE is pre-trained per fold on training users only.

C. Metrics and Statistical Tests

For the category task, we report macro-averaged F1 (macro-F1), which averages the per-class F1 so each of the seven categories counts equally; its reference point is the majority-class floor, the macro-F1 of always predicting the most common category. For the region task, we report accuracy at ten (Acc@10), the fraction of test visits whose true region appears among the model's ten highest-scoring guesses (a visit whose true region is absent from that fold's training data counts as an error); its reference point is the dedicated single-task model.

Where the joint model is meant to outperform, we test *superiority*. A *seed* is one complete repetition of the five-fold experiment, over the same folds, with a different random initialization; folds and seeds are the two axes of repetition. At four datasets, we use four seeds ($4 \times 5 = 20$ measurements) and pair the per-seed means (n=4 per dataset), with a Holm correction [29] across the comparisons; at Texas and California, measured with a single seed, we use a paired Wilcoxon signed-rank test over the five folds. Where it is meant only to stay even, we test *non-inferiority*, that it is no worse than the dedicated model by more than a two-point margin, with the two one-sided tests (TOST) procedure [30]. We fix the two-point margin in advance because a mobility-aware service acts on which region will be busy, not on a single rank position (Section III), so a two-point shift in Acc@10 is below the granularity at which such a service would behave differently. For scale, with 520 to 8,501 regions, a random top-ten guess includes the true region at most about two percent of the time. The equivalence is well powered: the paired difference has a small standard deviation (0.04 to 0.15 points at the four datasets with four seeds), so the procedure has power near 1.0 to declare equivalence when the true difference is near zero; the reported intervals would in fact pass a margin as small as half a point at Alabama and Arizona (Section VI-B).

D. Baselines

The comparisons play three roles. The first role is the per-task state of the art, on our data, under user-disjoint five-fold protocols. For next-category, this is POI-RGNN [31], re-implemented from its published architecture and hyperparameters, and a Markov baseline over recent categories, reporting the strongest order per dataset (Markov-K, up to the nine-visit window). For next-region, this is HMT-GRN [17], our primary comparison, an end-to-end recurrent model (trained as one piece from the raw check-ins) evaluated on the same data, folds, and initialization; we keep its shared multi-task skeleton, add a region-transition prior built per fold from training data, and drop its graph components and hierarchical beam search, which serve the next-place head that we do not predict. It is a region-native model, not a reproduction of the complete published system. We also run STAN [13], re-implemented from its published architecture and trained from the raw check-ins with its own embeddings and sequence construction under one fixed configuration, and a ReHDM [32] reference reported under its own published protocol.

The second role is two representation controls attributing the category gain to our specific design, not to contextualization in general or extra features: CTLE [11], the closest prior contextual check-in embedding, run end to end at Florida and with fixed weights at Alabama, Arizona, and Istanbul; and a feature-concatenation control appending raw per-visit features to a standard place embedding under the same model.

The third role compares the two ways of coupling the tasks. The dominant published coupling is the directed cascade, predicting the category first and conditioning the place prediction on it [22], [24]; CSLSL [23], its current form, reports the chain outperforming a parallel variant sharing one trunk. We test the cascade inside our own model rather than re-implementing those systems: we remove the shared trunk, where the two streams exchange information, and feed the predicted category forward into the region pathway, keeping the representation, outputs, and training configuration unchanged. This isolates the coupling topology (chain versus parallel) at no additional parameter or compute cost: the chain form drops the shared trunk and adds a conditioning projection of about 1,000 parameters. The standard place-level embedding [8] isolates what the per-visit representation adds. All other choices, including the nine-visit window, were fixed during development; the full training configuration for every model is in the released code (footnote 1).

VI. Results

A. The representation makes the category task learnable

Table II compares our check-in-level representation against a standard place embedding (HGI [8]) on next-category macro-F1. The comparison is like-for-like: target, single-task model, training configuration, windowing, and folds are identical; only the input representation changes (a vector per visit versus a vector per place), so the difference isolates the representation. The check-in-level column keeps one fixed configuration, not the per-dataset-tuned dedicated model of Table III. The check-in-level representation outperforms the place embedding by a wide gap at every state: $+29.31$ at Alabama, $+27.63$ at Arizona, $+39.62$ at Florida, $+37.95$ at California, and $+37.47$ at Texas. The same holds on Istanbul ($+28.09$). The gains fall in two bands, about $+28$ to $+29$ points at the smaller datasets (Istanbul, Alabama, Arizona) and about $+37$ to $+40$ at the three large states. A controlled test attributes most of that gain to the per-visit context itself: a per-place-averaged version of the same representation leaves roughly 64 to 90 percent of the gain (state-dependent) to the context that each visit carries, not to extra training signal.

The geometry of the vectors shows why the representation helps this task in particular: the per-visit vectors' silhouette score by category, a -1 to 1 measure of how tight and well separated the seven labeled groups are, is about 0.57, against about 0.00 for the place embedding. Their nearest-neighbor category purity, the share of nearest neighbors with the vector's own category, is about 0.98, against about 0.78 (both averaged over the five U.S. states). The same geometry does not separate regions, so the benefit is category-only (the joint model's spatial stream instead reads the region-level vectors of the same graph, Section IV-B). CTLE [11], the closest prior contextual embedding, also gives each visit its own vector, but it pretrains on place identifiers and timestamps alone, so the category vocabulary never enters its training, whereas our graph reads each visit's category and time of day as input features. Fine-tuned together with the task model at Florida at

TABLE I

DATASET STATISTICS, ORDERED BY REGION COUNT: FIVE GOWALLA STATES AND ONE INTERNATIONAL CITY (ISTANBUL, MASSIVE-STEPS). ALL SHARE THE SAME SEVEN ROOT CATEGORIES; THE REGION UNIT IS A CENSUS TRACT (A MAHALLE FOR ISTANBUL). WINDOWS ARE THE SLIDING NINE-VISIT INPUTS PLUS THE NEXT-VISIT TARGET; LENGTHS ARE PER-USER CHECK-IN COUNTS; DENSITY IS $100 \times$ CHECK-INS/(USERS $\times$ POIS); MAJORITY IS THE SHARE OF NEXT-VISIT LABELS IN THE MOST COMMON CATEGORY (FOOD IN EVERY DATASET).

Dataset	Source	Check-ins	Users	POIs	Regions	Windows	Max len	Avg len	Density (%)	Majority (%)
Istanbul	Massive-STEPS	462,615	23,694	29,816	520	271,666	817	19.5	0.065	33.4
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249	34.2
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145	34.0
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087	24.7
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066	31.0
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051	32.7

its authors' defaults, at the same 64 dimensions and windowing as ours, CTLE reaches 33.45 macro-F1 at its best epoch, one pass over the training data (29.69 at its final one), about two points below the place embedding under the same best-epoch rule and below our 75.15; frozen and fed to the same single-task model, it repeats the ordering at Alabama, Arizona, and Istanbul, with gaps of +37.8, +37.0, and +28.7 macro-F1. A feature-concat control (the place embedding joined with raw per-visit features, same model) does not close the gap either, so the gain comes from the hierarchical per-visit representation, not from contextualization alone or feature injection.

B. One model, two tasks

One model outperforms or matches the dedicated single-task models on both tasks. To calibrate the two scales: category macro-F1 spans seven classes (a majority-class floor of about 7%); region Acc@10 spans 520 to 8,501 regions, where a random top-ten guess is right at most about two percent of the time.

Table III reports the single joint model against the dedicated single-task ceilings (the level a joint model is usually expected at best to match). The dedicated category model is tuned per dataset over batch size and learning rate (the dedicated region models use the strongest fixed configuration). Throughout, each task is scored at its own validation-best epoch, under the same rule for the joint and dedicated models; a single jointly selected saved model per fold reproduces every joint result within 0.06 (category) and 0.11 (region) points, so no reported claim depends on reading two epochs.

Figure 3 plots the signed gains ordered by region count. On category, the joint model outperforms the dedicated ceiling at every dataset, by +5.34 to +9.40 macro-F1 (smallest at Florida, largest at Arizona, and +8.59 at Istanbul). On region (Acc@10), the joint model outperforms the dedicated ceiling at Florida +0.72 (4,703 regions), Texas +2.12 (6,553), California +2.20 (8,501), and Istanbul +0.28 (520), and stays a non-inferior match (TOST, ±2 pp) at Alabama (-0.31) and Arizona (+0.10). Across the five U.S. states, the region gain rises with the number of regions, from -0.31 at the smallest region count to +2.20 at the largest; region count and corpus size co-vary here, so we read the trend across the points rather than as a precise law.

Four of the six datasets, Alabama, Arizona, Florida, and Istanbul, are measured at four seeds for both models (20 measurements each), and the tests pair the per-seed means (n=4). On category, all four seeds favor the joint model at each dataset, and each gain is significant after a Holm correction across the four datasets (paired t, corrected $p < 0.001$). On region, all four are tested for equivalence (TOST, ±2 pp), and all four pass with 90% confidence intervals well inside two points: Alabama (-0.31; -0.48 to -0.14), Arizona (+0.10; +0.00 to +0.21), Florida (+0.72; +0.67 to +0.78), and Istanbul (+0.28; +0.24 to +0.32). At Florida and Istanbul, the whole interval lies above zero, so the joint model outperforms there as well, though both gains are smaller than the two-point margin of Section V-C; only Texas and California exceed that margin. At Arizona, the interval touches zero, so we report a match, not a gain. At Alabama, the whole interval lies below zero, a small but statistically significant deficit, still well within the two-point margin. At California and Texas, the dedicated ceilings average four seeds while the joint results are from a single seed over five folds. On those folds, all five favor the joint model on both tasks (one-sided paired Wilcoxon, p=0.031, the smallest value that five folds can produce), and the region gains have 90% intervals entirely above the two-point margin (+2.14 to +2.28 at California, +2.04 to +2.18 at Texas). More seeds at these two states are the remaining confirmation.

A reader used to multi-task learning [6] expects the harder task's training signal to drive the easier one's gain. As a control, we freeze the region pathway at the start of training so it can neither learn nor teach the category task, yet the full category gain survives at Alabama, Arizona, and Florida (within 0.3 of the joint model and far above the dedicated single-task model). We therefore attribute the category gain to a stronger shared trunk, obtained without a second model to serve (one model, one forward pass), not to the region task teaching the category one; we report this as a finding, not a hypothesis.

The joint model is also above every external baseline reported, on both tasks, at every dataset (Table III): the primary region-native comparison, HMT-GRN [17] (for example California 49.61 versus our 65.69), a STAN [13] trained from the raw check-ins and a ReHDM reference [32] on region, and POI-RGNN [31] and a Markov floor on category. These externals run on their own embeddings, so this comparison also includes the representation advantage of Section VI-A; the like-for-like anchor remains the dedicated column. A simple first-order Markov region floor reaches 43 to 65 Acc@10 across the datasets (computed under a non-overlapping windowing of the same data, indicative rather than protocol-matched); the joint model exceeds it by 12 to 23 points at all six datasets.

On principle, we prefer coupling the tasks in parallel rather than in a chain: neither target is subordinate to the other, so a chain must pick one to predict first and pass its errors to the other. The choice still had to be checked, because CSLSL reports its chain outperforming a shared-trunk parallel variant on its own benchmarks [23]. Rewired as a cascade (Section V-D), our model reaches the same combined score, the geometric mean of the two task metrics, as its parallel form at no additional parameter or compute cost: at Alabama, Arizona, Florida, and Istanbul, with both forms trained under the same configuration and initialization, the difference is about zero and neither task moves by more than a quarter of a point. We read this as a defense of the parallel design, not a claim that we outperform the cascade: on this representation the coupling topology makes no measurable difference, and the simpler parallel structure loses nothing. Two qualifications bound this reading: the cascade runs under the configuration tuned for the parallel model, and its form was fixed in advance, so tuning the cascade itself remains future work.

C. External validity (Istanbul)

The finding holds on a non-U.S. city under the same representation, sliding-window protocol, and training setup as the U.S. states. On Istanbul, the joint model outperforms the dedicated category ceiling by +8.59 macro-F1 (54.74 to 63.33) and is slightly above the dedicated region ceiling at +0.28 Acc@10 (75.16 to 75.44), a gain with statistical support (the whole 90% confidence interval of the paired difference lies above zero, and TOST, ±2 pp, also passes); both models average four seeds over five folds. The comparable quantity is the gain over the ceiling, not the absolute Acc@10, since region counts differ across datasets (520 mahalle here). The external baselines show the same ordering: on category, the joint model is above the Markov floor (24.55) and POI-RGNN [31] (30.12), and on region, it is above ReHDM [32] (69.33), STAN [13] (61.86), and HMT-GRN [17] (60.4). The U.S. result repeats on a different continent and region unit.

VII. Discussion and Limitations

One model serves both tasks: a shared trunk carries the semantic context that lifts the next-category task over a dedicated model at every dataset, and a private spatial path keeps the next-region task competitive, a non-inferior match (TOST, ±2 pp) at the two small U.S. states and a tested gain at the other four. Across the five U.S. states, the region gain rises with the number of regions (Figure 3). The joint model matches or outperforms the dedicated single-task model on both tasks (Table III).

A service could treat the model's top-ranked next regions as an anticipatory set: at California, ten regions of 8,501 contain

TABLE II
Next-category macro-F1 of the check-in-level representation against a standard place embedding (HGI), same single-task model, training configuration, and folds (mean and fold standard deviation, seed 0).

Dataset	Check-in level	Place level	Δ
Istanbul	**54.65** ±0.6	26.56 ±0.5	+28.09
AL	**55.87** ±2.4	26.56 ±1.0	+29.31
AZ	**57.13** ±1.3	29.50 ±1.0	+27.63
FL	**75.15** ±0.5	35.53 ±0.3	+39.62
TX	**69.95** ±0.2	32.48 ±0.6	+37.47
CA	**70.26** ±0.3	32.31 ±0.4	+37.95

The matching place-level value at Alabama and Istanbul (26.56) is a coincidence of two independent runs; their per-fold values differ.

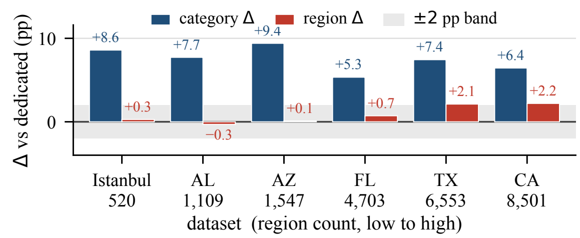

Fig. 3. Signed gains of the joint model over the dedicated single-task models, ordered by region count. The category gain (macro-F1) is positive at every dataset; the region gain (Acc@10) rises across the five U.S. states (TX and CA: one seed) and is also positive at Istanbul. The ±2-point band applies to region only.

the true next region 65.69 percent of the time, over 500 times better than picking ten at random. On the four datasets that we measured (Alabama, Arizona, Florida, and Istanbul; a single seed over five folds), the ten shortlisted regions are about 3 to 8 kilometers from the shortlist's center (medians from region centroids), against 20 to 241 kilometers between two regions drawn at random from the same map. This remains motivation, not a measured service result.

Three limits qualify these results. First, the joint numbers at California and Texas use a single seed over five folds, while the other four datasets average four seeds for both models. Second, our representation is trained once over all places; visits to places never seen during training are the single effect that we cannot fully isolate. A planned follow-up trains it on each fold's training places only and embeds unseen visits. Third, although a mobility-aware service motivates this work, we do not build or evaluate one.

Our representation is a fixed per-visit vector that any downstream model can consume; structural check-in analysis has named machine learning as a next step [5], and this study moves in that direction.

VIII. Conclusion

We tested whether one model can predict both the category and region of a user's next visit. First, a check-in-level representation, one vector per visit rather than one per place, makes the next-category task far more learnable than the standard place-level one. Second, on that representation, a single multi-task model outperforms a dedicated category model at every

TABLE III

ONE MODEL, TWO TASKS: THE SINGLE JOINT MODEL AGAINST THE DEDICATED SINGLE-TASK MODELS AND THE EXTERNAL BASELINES, ORDERED BY REGION COUNT. CATEGORY IS MACRO-F1, REGION IS ACC@10. BOLD MARKS A STATISTICALLY SUPPORTED IMPROVEMENT OVER THE DEDICATED MODEL; IN THE REGION COLUMNS, $\uparrow$ MARKS THAT SUPPORTED IMPROVEMENT AND $\approx$ A NON-INFERIOR MATCH (TOST, ± 2 PP; SECTION VI-B).

		Next-category (macro-F1)				Next-region (Acc@10)				
Dataset	**Regions**	**Markov-K**	**POI-RGNN**	**Dedicated**	**Joint (ours)**	**HMT-GRN**	**ReHDM**	**STAN**	**Dedicated**	**Joint (ours)**
Istanbul	520	24.55	30.12	54.74 $\pm$0.09	**63.33** $\pm$0.02	60.4	69.33	61.86	75.16 $\pm$0.01	**75.44** $\pm 0.04^{\uparrow}$
AL	1,109	20.50	23.80	56.82 $\pm$0.03	**64.54** $\pm$0.10	57.05	65.38	60.72	70.11 $\pm$0.10	69.80 $\pm 0.05^{\approx}$
AZ	1,547	23.92	27.64	56.43 $\pm$0.10	**65.84** $\pm$0.02	43.70	53.00	49.86	59.46 $\pm$0.08	59.56 $\pm 0.05^{\approx}$
FL	4,703	29.74	34.49	74.51 $\pm$0.03	**79.85** $\pm$0.03	63.74	64.49	72.99	76.70 $\pm$0.02	**77.42** $\pm 0.03^{\uparrow}$
TX	6,553	28.67	33.03	69.79 $\pm$0.08	**77.23**$^{\circ}$ $\pm$0.12	53.85	48.81‡	61.67†	64.95 $\pm$0.01	**67.07**$^{\circ}$ $\pm 0.45^{\uparrow}$
CA	8,501	27.58	31.78	70.60 $\pm$0.07	**77.04**$^{\circ}$ $\pm$0.20	49.61	50.26‡	58.52†	63.49 $\pm$0.02	**65.69**$^{\circ}$ $\pm 0.30^{\uparrow}$

HMT-GRN (region-native, on our folds, scored on visits whose region appears in training) is the primary external comparison; STAN (our re-implementation) and ReHDM (its own protocol) are references. Markov-K reports the strongest order per dataset. †STAN partial folds: Texas four of five, California two of five (seed 0). ‡ReHDM at Texas and California: a single seed. $^{\circ}$Joint entries at California and Texas: a single seed over five folds; all other entries average four seeds. $\pm$: sd across seeds, or folds for single-seed entries.

dataset and a dedicated region model at four of the six datasets, remaining non-inferior (TOST, ± 2 pp) at the other two. One model with a shared semantic context and a private spatial path anticipates both what a user will do next and where.

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